

# Histidine-Rich Glycoprotein Modulation of Immune/Autoimmune, Vascular, and Coagulation Systems

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**Abstract** Histidine-rich glycoprotein (HRG) is plasma glycoprotein that has a multidomain structure, interacts with many ligands, and exhibit many modulatory functions in diverse biological systems. HRG ligands include  $\text{Zn}^{2+}$ , tropomyosin, heparin and heparan sulphate, plasminogen, plasmin, fibrinogen, thrombospondin, IgG, FcγR, and complement. In many cases, the histidine-rich region of the molecule enhances ligand binding after interaction with  $\text{Zn}^{2+}$  or exposure to low pH, conditions associated with sites of tissue injury or tumor growth. The multidomain nature of HRG and diverse ligand binding properties indicates that it can act as an extracellular adaptor protein, bridging together different ligands mainly on cell surfaces. Apart from cell surface molecules, HRG can differentially target IgG, preventing generation of insoluble immune complexes. HRG binds to most cells primarily via heparan sulphate proteoglycans. HRG can enhance clearance of apoptotic and necrotic phagocytes as well as immune complexes. The anti-angiogenic properties of HRG are well established in vitro and in vivo. HRG can modulate other physiological processes such as cell adhesion and migration, fibrinolysis and coagulation, complement activation. This review presents an

update on the molecular, structural, biological, and clinical properties of HRG and associated autoimmunity conditions.

**Keywords** Histidine-rich glycoprotein · Immune complex · Thrombospondin · Clearance · Protective molecule · Lupus · Rheumatoid arthritis · Angiogenesis · Autoimmunity

## Introduction

Histidine-rich glycoprotein (HRG) is a heparin-binding plasma protein 67- to 75-kDa protein, composed of 507 amino acid residues, present on the plasma at concentration of ~100 μg/ml [1, 2]. HRG is present in the blood of vertebrates, including humans and some invertebrates [1, 2]. In mammals, the majority of HRG is synthesized by the liver, found in the plasma, stored in the α-granules of thrombocytes, and is released upon thrombin activation [3]. Structurally, the HRG molecule consists of three main domains: (a) The NH<sub>2</sub>-terminal domain contains two cysteine protease inhibitor (cystatin)-like stretches, which allows the classification of HRG as a member of the cystatin superfamily together with α<sub>2</sub>HS-glycoprotein, cystatin, and kininogen. (b) The central domain is very rich in proline and histidine residues, and the human form contains multiple GHHPH pentapeptide tandem repeats. The central domain and the COOH-terminal domain are disulfide bonded to the cystatin-like stretched in the NH<sub>2</sub>-terminal domain [4]. This specific architecture of the HRG molecule facilitates several functions by simultaneous interaction with numerous ligands. Previous studies have shown that HRG binds to heparin, heparan sulfates, thrombospondin (TSP), fibrinogen, plasminogen, certain complement proteins, divalent metal ions, and immunoglobulins [1, 4–6]. Based on the multiple ligand binding potential, HRG binds

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to a wide range of cells such as endothelial cells, fibroblasts, T cells, macrophages, erythrocytes, platelets, monocytoid cell lines, and necrotic cells [5, 7–11]. The receptors for HRG on these cells have not been finally identified; however, inhibition studies suggest that HRG binds to TSP, FcR, and glycosaminoglycans [5, 9, 10]. The multi-pattern binding properties of HRG enable its diverse biological activities. The various functions of HRG molecule known so far encompass modulation of fibrinogenesis [12], inhibition of insoluble immune complex formation [13], enhancement of apoptotic cells clearance by macrophages [14, 15], anti-angiogenic activities [16–18], and antibacterial activity [19]. This review will focus on the biological activities of HRG and its association with autoimmunity.

### HRG as a Protective Molecule

#### HRG Binds to Human C1q and IgG and Inhibits Insoluble Immune Complexes Formation

Certain autoimmune diseases such as systemic lupus erythematosus (SLE) or rheumatoid arthritis are associated with the production of autoantibodies [20, 21]. Based on in vitro studies, autoantibodies can be cross-linked with their cognate antigens to form insoluble immune complexes (IIC), which can precipitate on the tissues (e.g., kidney or joint synovial membrane) and lead to a lack of molecular transport and consequent tissue injury. Evidence suggests that the complement components C3b and C1q can modulate formation of IIC in vitro. C1q is known to enhance the formation of IIC by binding to IgG molecules via the C2 domain and may lead to the cross-linking of IgG molecules. Studies addressing the effect of HRG and C1q on the formation of IIC between ovalbumin and polyclonal rabbit anti-ovalbumin IgG revealed that physiological concentrations of HRG bind strongly to C1q and to IgG, inhibiting the formation of IIC in vitro [6]. C1q and IgG binding region on HRG are located in the N-terminal fragment. HRG bind differentially immunoglobulin subclasses [22]. HRG bind to the kappa L-chain containing IgG1 and IgG2 (IgG1kappa and IgG2kappa) in approximately four and approximately tenfold faster than that for the binding of HRG to lambda L-chain containing IgG1 and IgG2 (IgG1lambda and IgG2lambda), respectively. In contrast, the binding of HRG to IgG3kappa and IgG4kappa is 9- and 20-fold slower than that for the binding of HRG to IgG3lambda and IgG4lambda, respectively. The binding of HRG to immunoglobulins containing the kappa L-chain (particularly IgG1kappa), but not lambda L-chain, is potentiated in the presence of a physiological concentration  $Zn^{2+}$ .

Based on the results that HRG can bind to IgG and inhibit the IIC, it was of interest to determine whether HRG

can inhibit the formation of IIC formed by interaction of rheumatoid factor (RF) with human IgG-containing IC. Employing light scattering techniques, the effect of HRG on the formation of IIC between RF and IC containing human IgG was tested in three different models [23]. In all three models, physiological concentrations of HRG could block the formation of IIC induced by RF. Optical biosensor studies of the RF-IgG interaction also revealed that HRG can mask the epitopes on IgG recognized by RF, therefore, block the RF binding to ICs [23]. Furthermore, HRG can partially solubilize IIC [23]. These data indicate that HRG can regulate the formation of IIC induced by RF at three levels: namely, by inhibiting the initial recognition of IgG containing IC by RF, by inhibiting the subsequent IIC of IgG containing IC by RF, and by solubilizing already formed IIC. These findings raised the possibility that HRG may be an important inhibitor of the formation of pathogenic IC diseases such as SLE and RF.

#### HRG Regulates the Binding of Monomeric IgG and Immune Complexes to Monocytes

HRG can interact with FcγRI and block monomeric IgG binding to monocytes. When incorporated in IgG containing IC, HRG can enhance the uptake of IC by monocytes, probably via its heparan sulfate-binding domain.

It is well established that monomeric IgG binds to FcγRI on mononuclear cells and HRG interacts with mouse peritoneal macrophages [24]. HRG differentially regulates the functions of FcγR subclasses. For example, HRG reduced the efficiency of FcγRII (Fcγ2b/γ1R)-dependent phagocytosis of erythrocytes conjugated with monoclonal IgG2b or IgG1 by macrophages pretreated with HRG for 24 h. However, when similar studies were performed using erythrocytes coated with monoclonal IgG2a, HRG was less effective in inhibiting FcγRII (Fcγ2b/γ1R)-mediated phagocytosis [18]. HRG bind FcγRI expressed on human monocytic cell lines THP1 and U937 [9]. This claim is proven by inhibition of HRG binding to THP1 by anti-FcγRI Abs and the lack of IgG interference with the binding of HRG to the T cell lines lack FcγRI [9]. Pretreatment of THP1 cells with HRG at physiological concentration significantly inhibit the binding of monomeric human IgG to THP1 cells via masking the IgG binding site on the FcγRI. Addition of IC to HRG-THP1-IgG cross talk did not change the ability of HRG to inhibit the binding of monomeric IgG to THP1 cells. However, the fact that pretreatment of the cells with HRG did not affect the binding of IC to these cells point to the possibility that the IC bind low affinity FcR not recognized by HRG. Interestingly, pre-incubation of monomeric IgG with HRG enhance the IgG targeting THP1 cells. The incorporation of HRG into IC may increase the avidity of these IC for the

cell surface because of the formation of multivalent interactions between IC and heparan sulfate receptors for HRG. It should be noted that the molar ratio of HRG/IgG is an important factor in determining whether HRG potentiates (HRG/IgG molar ratio ~1–10:1) or inhibits (HRG/IgG molar ratio=50:1) the binding of free IgG to FcγRI. In the normal physiological conditions, plasma HRG (2 μM) will not enhance the binding of free IgG (~60 μM) to monocytes, as the molar ratio of HRG/IgG is ~1:30, and the most HRG would be associated with soluble IgG and unavailable to interact with FcγRI [9].

#### HRG Enhance the Clearance of Apoptotic Cells

During the last decade, bunch of evidence prove a defect in clearance of apoptotic cells in SLE progression [25–27]. It was proposed that long circulating apoptotic cell, which expose nuclear contents to the immune system, followed by generation of neoantigens as a result of re-clustering of the internal antigens and post-translation modification processes all are the source of more than 100 autoantigens in lupus [20, 21]. Because of the brakeage of self tolerance, autoimmune processes are switched on and may induce autoimmune state. Therefore, efficient masking of autoantigen epitopes and effective clearance of apoptotic cells by phagocytes are essential to prevent autoimmune reactions. Several receptors and ligands are known to serve as mediators for the uptake of apoptotic cells by macrophages, some dependent on the direct interaction of apoptotic cells with phagocyte receptors such as CD14, phosphatidylserine, and scavenger receptors, in association with cytoskeletal proteins such as ABC1 cassette transporters, dock 180, and Wiskott–Aldrich syndrome protein on phagocytes. Other ligand binding require an intermediate molecule between the circulating protein and the cells, for example, the acute phases C-reactive protein (CRP) at high levels, engages with phosphatidylserine on apoptotic cells and activates the classical complement cascade, resulting in the release of C1q and deposition of C3b on apoptotic cells. Additional example is the high concentration of secreted forms of TSP which potentiate uptake by bridging apoptotic cells and the  $\alpha_v\beta_3$ /CD36 complex on macrophages [28]. HRG specifically bind to apoptotic T cells and to mature human monocyte-derived macrophages (HMDM) and potentiate the ingestion of apoptotic cells [14]. The HRG-dependent mechanism mediates removal of apoptotic cells under normal conditions, whereas in pathological conditions such as acute or chronic inflammatory disorders, TSP and CRP/complement-dependent mechanisms also contribute. Gorgani et al. [14] show that purified HRG increase the number of human HMDM-containing apoptotic cells and accelerate the ingestion. HRG potentiates phagocytosis of apoptotic cells through bridging naked DNA on apoptotic cells with FcγRI on mature macrophages [14]. Anti-FcγRI

mAb inhibit HRG binding to HMDM, whereas naked DNA inhibit HRG binding to apoptotic cells. Thus, anti-FcγRI or DNA abrogate the HRG-dependent ingestion. These findings indicate that HRG, by acting as a bridge between DNA on apoptotic cells and FcγRI on HMDM, is an important mediator for the clearance of apoptotic cells by macrophages.

It can be summarized that HRG-dependent clearance mechanisms are active earlier than other systems such as CRP or TSP, as HRG, both at acute phase and physiological concentrations, can potentiate apoptotic cell uptake by macrophages, whereas CRP or TSP-mediated mechanisms are activated only at acute phase [14]. In parallel, Jones et al. [29] showed that HRG can link necrotic cells to monocytes via cytoplasmic ligands in heparin-sulfate-independent manner exposed on necrotic cells. This interaction is mediated by the amino-terminal domain of HRG. It is important to notice that circulating HRG level is significantly lower in lupus patients [30], suggesting active involvement of HRG during acute as well as chronic inflammation. One may speculate that low levels of HRG in lupus plasma may explain partially the decreased clearance potential of phagocytic cells.

#### HRG Modulation of Angiogenesis

*HRG as anti-angiogenesis molecule* Angiogenesis is critical for the growth and proliferation of tumors as well as for normal development such as wound healing, placenta development menstrual cycle [31]. In normal physiological condition, the vascular-bed is regulated by pro- and anti-angiogenic factors. Prolonged and excessive angiogenesis is implicated in a number of pathological conditions such as rheumatoid arthritis, diabetic retinopathy, and tumor growth and metastases. The vasculature is regulated by pro- and anti-angiogenic factors such as fibroblast growth factor (FGF) and vascular endothelial growth factor (VEGF), stimulators of endothelial cells in vitro and in vivo. To date, number of self endogenous factors serve as negative regulators for angiogenesis (e.g., plasma proteins, basement membrane proteins, and serine protease inhibitors-serpins), all inhibit endothelial cell chemotaxis in vitro and in vivo [32].

Zhang et al. [33] demonstrate the anti-angiogenic activity of HRG in vitro. Others show rabbit-HRG-mediated inhibition of endothelial cells tube formation stimulated by FGF-2 or VEGF on a Matrigel surface as well as cell proliferation of FGF-2-stimulated endothelial cells [16]. Using proteolytic fragments of HRG, it could be demonstrated that the anti-angiogenic activity of HRG is localized in the His/Pro-rich domain and further confirmed in two in vivo models of angiogenesis: the chorioallantoic membrane of the chick assay and the mouse Matrigel plug assay. Caspase-3 activation is observed in endothelial cells stimulated with FGF-2 in the presence of HRG, suggesting

that apoptosis of activated endothelial cells may be one of the mechanisms underlying HRG anti-angiogenic activity. Finally, the His/Pro-rich domain of HRG reduce tumor cell number when tumor cells are co-inoculated in the Matrigel plug assay [16]. Olsson et al. [34] constructed recombinant HRG with anti-angiogenic potential. This group define the HRG peptide derivative of 35 aa named HRGP330, derived from the His/Pro-rich domain of HRG, as the minimal anti-angiogenic active HRG fragment. By use of a signal transduction antibody array representing 400 different signal transduction molecules, this group shows that HRGP and the synthetic peptide HRGP330 specifically induce tyrosine phosphorylation of focal adhesion kinase and its downstream substrate paxillin in endothelial cells [35]. HRGP or HRGP330 treatment of endothelial cells induce disruption of actin stress fibers, a process reversed by treatment of cells with the FAK inhibitor geldanamycin. In addition, HRG and HRGP330 inhibit VEGF-mediated endothelial cell tubular morphogenesis in a three-dimensional collagen matrix. In contrast, VEGF-induced proliferation was not affected by HRG or its synthetic derivative. These data show that HRGP targets focal adhesions in endothelial cells, thereby disrupting the cytoskeletal organization and the ability of endothelial cells to assemble into vessel structures [35].

**HRG as pro-angiogenesis molecule** HRG binds with high affinity to thrombospondin-1 (TSP-1), a homotrimeric glycoprotein that is a potent inhibitor of angiogenesis [5]. The anti-angiogenic activity of TSP-1 is mediated by the binding of properdin-like type I repeats to the receptor CD36. The binding of HRG to TSP-1 is mediated by TSP-1 repeats. HRG is co-localized with TSP-1 in the stroma of human breast cancer specimens, and this interaction mask the anti-angiogenic epitope of TSP-1. In vitro studies reveal that HRG inhibit endothelial cell migration and tube formation and in vivo corneal angiogenesis. In-contrast, HRG may inhibit the anti-angiogenic effect of TSP-1. These studies suggest that HRG can modulate the anti-angiogenic activity of TSP-1 and identify a potential mechanism of resistance to the anti-angiogenic effect of TSP-1. This kind of HRG angiogenesis modulation has a vast contribution to the regulation of the vascular-bed in diverse clinical states.

### HRG as a Modulator of Hemostasis

Plasma HRG circulates as a complex bound to the lysine-binding site of plasminogen, and this serves to reduce the effective plasminogen concentration, inhibit binding of plasminogen to fibrin/fibrinogen, and retard fibrinolysis in

vitro [36]. High plasma levels of HRG or HRG deficiencies are associated with thrombophilia [37].

Construction of the HRG-deficient mice enabled to study the physiological function of HRG [38]. These mice have no immunologically detectable HRG protein in their plasma, and the mice are fertile; prothrombin time is significantly shorter, and reconstitution of the plasma samples with purified human HRG normalize the prothrombin time. Thus, HRG may regulate the extrinsic pathway of coagulation in mice by interacting with thrombin. In addition, the HRG-deficient plasma had higher anti-thrombin activity, suggesting an additional and alternative mechanism of HRG anticoagulant activity. Additional aspect in which HRG contribute to homeostasis address the thrombin system. Anti-thrombin is a potent physiological negative regulator of the blood coagulation inhibiting thrombin, factor Xa and factor Ixa. Its inhibitory activity towards serine proteases is much enhanced in the presence of heparin. HRG has a high binding affinity for heparin, and therefore, potentially competes with the binding of heparin to anti-thrombin. Thus, HRG inhibits the high-affinity binding of anti-thrombin to thrombin as well as the anti-factor Xa activity of anti-thrombin. Therefore, HRG is considered a modulator of anticoagulant activity of heparin in human plasma [39]. The authors suggested that HRG can antagonize anti-thrombin activity in vivo.

Apart from plasmatic coagulation, hemostasis can also be modulated on the level of platelets, which are known to carry HRG [3]. When the effect of HRG deficiency on platelet function was analyzed, the bleeding time in HRG<sup>-/-</sup> mice was significantly shorter than in HRG<sup>+/+</sup> mice. This result suggested that in addition to its influence on plasmatic coagulation, HRG inhibits platelet function. HRG strongly binds platelets in the presence of Zn<sup>2+</sup> and inactivates platelet thrombospondin, a strong promoter of platelet aggregation [5]. These findings suggest that HRG plays a role as both an anticoagulant and an anti-fibrinolytic activities and may regulate platelet function in vivo.

### Conclusion Remarks

We present herein, a rainbow of HRG contributions to regulation of diverse immunological/biological activities, being a protecting molecule on one hand and modulator molecule of homeostasis on the other hand. Because of the ability of HRG to target different ligands as soluble molecules or as a cell surface epitopes and based on its multidomain architecture, HRG can immunomodulate many pathological conditions. The role of HRG in autoimmunity may include attenuation of IIC and IgG–RF generation, enhance clearance of IC, and the clearance of apoptotic and necrotic cells, conditions associated with SLE



and rheumatoid arthritis. Apart of these biological functions, HRG and its peptidic derivative have anti-angiogenic properties, inhibiting endothelial tubular formation, inhibiting tumor proliferation and induction of apoptosis in tumor cells in vitro and in tumors in animal models. Therefore, HRG and its synthetic derivative may be apply at first for treatment of several models of autoimmune diseases as well as diverse tumor models.

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